

Structure-Preserving Semismooth Newton Method for Large-Scale Hybrid AC/DC Power Flow With Converter Limit-Induced Mode Switching

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Abstract—Converter-dominated hybrid ac/dc grids exhibit operating-mode changes when voltage-source converters reach ac-current, active-power, reactive-power, or dc-voltage-control limits. Conventional power-flow implementations commonly handle these events by external if-else switching, which changes equation semantics and may rebuild the sparse Newton system. This paper develops a fixed-dimension complementarity formulation and a structure-preserving semismooth Newton method for balanced positive-sequence hybrid ac/dc power flow. Each limited converter owns six explicit local variables representing ac active and reactive power, dc power, internal-voltage components, and a complementarity multiplier. The formulation covers current-magnitude scaling, active-or reactive-power priority, saturated dc-voltage droop, and a grid-forming Norton source behind virtual impedance. An exact local-block Schur elimination removes the converter variables before sparse network factorization without forming dense inverses. Local reciprocal-condition and reduced/full-system backward-error certificates guard every Schur step and trigger a complete sparse-LU fallback when necessary. Conditional local superlinear and quadratic convergence results are stated under BD-regularity and strong semismoothness, while runtime diagnostics deliberately certify only the selected generalized-Jacobian element. Preliminary production results on hybridized case300 and ACTIVSg2000 systems show exact full/Schur root agreement within 10^{-8} p.u. The reduced formulation decreases dimension from 4054 to 4006 on ACTIVSg2000 and lowers median linear-solve and end-to-end times by 14.0% and 12.5%, respectively. The same method is exercised under simultaneous converter limiting, degenerate intersections, weak grids, infeasible operating points, grid-forming islands, OpenDSS circuit replay, optimal-power-flow replay, and transient initialization.

Index Terms—Complementarity constraints, grid-forming converter, hybrid ac/dc power flow, mode switching, semismooth Newton method, Schur complement, voltage-source converter.

I. Introduction

HYBRID ac/dc grids are increasingly operated through voltage-source converters rather than through passive interfaces. Multi-terminal dc links, offshore-wind collection systems, converter-interfaced generation, and grid-forming resources make converter control limits part of the network equilibrium itself. A converter may regulate active and reactive power, dc voltage, or an internal ac voltage during normal operation, but semiconductor current limits remain hard physical constraints. Once a limit becomes active, the unconstrained command can no longer be maintained and the steady or quasi-steady operating point changes control regime.

This limit-induced mode switching is more involved than the classical generator PV/PQ transition. A converter can encounter an ac-current disk, a dc power bound, saturated dc-voltage droop, and a selected priority between active and reactive current. For a grid-forming converter, current limiting also modifies the relation between the internal voltage source and the terminal voltage. Consequently, a post-limit equilibrium may require explicit internal voltage and current variables rather than a bus-level net-injection correction. These issues are especially important in islanded networks, where a fixed GFM internal phasor can provide the absolute angle reference without manufacturing a terminal slack bus.

Unified hybrid ac/dc power-flow methods have improved the treatment of ac networks, dc networks, and converter coupling [1]–[3]. In parallel, GFM control and fault-current-limiting studies have clarified how a voltage-source behavior is modified by current constraints [4], [5]. Most power-flow implementations nevertheless retain an external engineering loop of the form “solve, inspect a limit, switch the device model, and solve again.” This approach is practical for isolated limits, but its equation dimension, variable semantics, and sparse pattern may depend on the switching sequence. Simultaneous activation, priority ties, and return switching can also make the final result sensitive to implementation order.

Complementarity equations provide a fixed-dimensional alternative. A limit and its multiplier are solved together, allowing the inactive and active regimes to be represented by one nonsmooth system [6], [7]. Semismooth Newton methods can then achieve fast local convergence without enumerating all mode combinations [8]. Their use in a production hybrid ac/dc solver, however, raises three additional questions. First, the converter must own explicit local variables so that device equations and result certificates do not have to be reconstructed from aggregated bus injections. Second, the fixed equation layout must translate into a measurable sparse-linear-algebra advantage rather than only a software convenience. Third, local convergence claims must distinguish a numerically accepted generalized-Jacobian element from the stronger BD-regularity condition over the B-subdifferential.

This paper addresses these questions through a balanced positive-sequence hybrid ac/dc formulation with explicit

converter blocks. Each supported VSC owns the local state

$$z_c = (P_{ac,c}, Q_{ac,c}, P_{dc,c}, E_{r,c}, E_{i,c}, \lambda_c), \quad (1)$$

where the internal-voltage slots are active for GFM operation and are fixed by identity equations for power-port modes. Current-magnitude scaling uses a Fischer–Burmeister equation. Active- and reactive-power priorities use nested median projections. Vdc-Q operation includes a saturated dc-voltage droop command, and GFM operation uses an internal voltage behind virtual impedance. All equations remain present across limit activation.

The main contributions are as follows.

- 1) A unified fixed-dimensional converter-limit model is developed for PQ, Vdc-Q, and GFM operation. It combines the ac-current disk, magnitude or P/Q priority, converter energy balance, saturated dc-voltage droop, and GFM Norton equations while preserving stable device identity and explicit local result variables.
- 2) A production semismooth Newton method is formulated with analytic network/local generalized Jacobians. Exact Fischer–Burmeister and median equations define the reported root; optional Kanzow/CHKS smoothing provides a fixed-pattern continuation for difficult degenerate points. A full sparse Newton fallback remains available at every iteration.
- 3) An exact local-block Schur method eliminates all six-state converter blocks before network factorization. The method has $O(m)$ local dense work for m converters, removes exactly $6m$ unknowns, and reduces the assembled structural nonzeros by at least $48m$ under the stated three-port coupling pattern. Local reciprocal-condition and normwise backward-error tests guard the reduced and reconstructed full steps.
- 4) The convergence statement separates mathematics from runtime evidence. BD-regularity plus semismoothness yields local superlinear convergence, and strong semismoothness yields the standard quadratic rate. Runtime certificates establish nonsingularity and backward stability only for the selected generalized-Jacobian element; they are not presented as a proof over every element of the B-subdifferential.
- 5) The implementation is exercised through registered production tests, OpenDSS circuit validation, OPF-to-PF replay, transient Norton initialization, simultaneous limits, weak-grid and infeasible cases, GFM islands, and real hybrid case300/ACTIVSg2000 benchmarks. The preliminary results also retain a negative small-system performance result, which motivates an explicit and configurable production admission rule.

Figure 1 summarizes the problem, model ownership, and solver structure. The central distinction is that limit activation changes residual values and generalized derivatives, but not the converter state, equation coordinates, or sparse ownership map.

The paper is organized as follows. Section II defines the network and converter equations. Section III presents the semismooth Newton method. Section IV derives the exact local-block Schur elimination and its cost. Section V states the regularity and local-rate conditions. Section VI reports the current production evidence and its limitations.

II. Hybrid AC/DC and Converter-Limit Model

A. Network variables and nodal balances

Let $\mathcal{N}_{ac}$ and $\mathcal{N}_{dc}$ denote the energized ac and dc buses, and let $\mathcal{C}$ denote the supported limited VSCs. The network state is

$$x = (\theta, V, V_{dc}), \quad (2)$$

with only the coordinates required by the selected bus equations retained. In a conventional grid-connected component, one terminal angle is omitted or fixed. In a slackless GFM island, all terminal voltage magnitudes and angles remain variables; a fixed internal GFM phasor supplies the absolute reference through the device equation.

For ac bus i , the calculated complex injection is

$$S_i^{\text{net}}(V, \theta) = V_i e^{j\theta_i} \sum_{k \in \mathcal{N}_{ac}} Y_{ik} V_k e^{j\theta_k}. \quad (3)$$

The active- and reactive-power residuals are written in the production “specified minus calculated” convention,

$$F_{P,i} = P_i^{\text{spec}}(x, z) - \Re(S_i^{\text{net}}), \quad (4a)$$

$$F_{Q,i} = Q_i^{\text{spec}}(x, z) - \Im(S_i^{\text{net}}). \quad (4b)$$

For dc bus j , with conductance matrix G_{dc} ,

$$F_{dc,j} = P_{dc,j}^{\text{spec}}(x, z) - V_{dc,j}(G_{dc}V_{dc})_j. \quad (5)$$

Each converter contributes its explicit $(P_{ac,c}, Q_{ac,c}, P_{dc,c})$ directly to the corresponding nodal balances. This ownership is important: converter power, limit activity, and internal-voltage results are solver variables and certificates, not values inferred after convergence from a bus net injection.

B. Fixed local converter state

Every $c \in \mathcal{C}$ owns the six-state vector in (1). The complete nonlinear system is

$$H(w) = 0, \quad w = (x, z_1, \dots, z_m), \quad m = |\mathcal{C}|, \quad (6)$$

where H stacks the network balances and six local equations per converter. The local state has one meaning across steady power flow, quasi-steady replay, OPF post-processing, and transient initialization. For non-GFM power-port modes, $E_{r,c} = E_{i,c} = 0$ are identity equations; the slots are retained so the dimension and sparse coordinates do not change with control role.

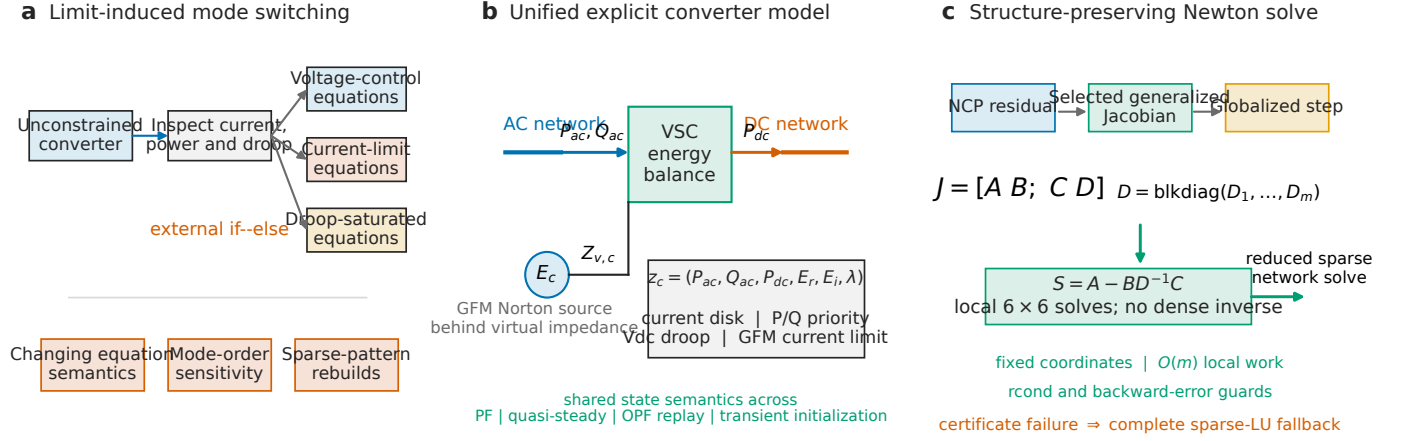


Fig. 1. From limit-induced switching to the proposed fixed-dimensional framework. (a) External mode switching can change equation semantics and sparse structure. (b) Each VSC instead owns one explicit six-state block, including the GFM internal voltage behind virtual impedance. (c) The resulting fixed generalized-Jacobian pattern supports certified local-block Schur elimination with complete sparse-LU fallback.

C. Power command and dc-voltage droop saturation

For PQ operation,

$$P_c^{\text{ref}} = P_c^{\text{set}}, \quad Q_c^{\text{ref}} = Q_c^{\text{set}}. \quad (7)$$

For Vdc-Q operation, the active-power command is

$$P_c^{\text{raw}} = k_{v,c} (V_{\text{dc},c}^2 - (V_{\text{dc},c}^{\text{set}})^2), \quad (8a)$$

$$P_c^{\text{ref}} = \text{clip}(P_c^{\text{raw}}, P_c^{\text{min}}, P_c^{\text{max}}), \quad (8b)$$

$$Q_c^{\text{ref}} = Q_c^{\text{set}}. \quad (8c)$$

Inside the open unsaturated interval, $\partial P_c^{\text{ref}} / \partial V_{\text{dc},c} = 2k_{v,c} V_{\text{dc},c}$; outside it the derivative is zero. The power bounds are therefore represented inside the local equations rather than through an external mode switch.

D. Current disk and power-port priorities

For terminal voltage magnitude V_c and current limit I_c^{max} , define

$$R_c = V_c I_c^{\text{max}}, \quad g_c = R_c^2 - P_{\text{ac},c}^2 - Q_{\text{ac},c}^2. \quad (9)$$

The magnitude-priority model radially scales the requested complex power:

$$(1 + \lambda_c) P_{\text{ac},c} - P_c^{\text{ref}} = 0, \quad (10a)$$

$$(1 + \lambda_c) Q_{\text{ac},c} - Q_c^{\text{ref}} = 0, \quad (10b)$$

$$\phi_{\text{FB}}(\lambda_c, g_c) = 0, \quad (10c)$$

where

$$\phi_{\text{FB}}(a, b) = \sqrt{a^2 + b^2} - a - b. \quad (11)$$

Equation (11) is zero exactly when $a \geq 0$, $b \geq 0$, and $ab = 0$.

For P-first and Q-first operation, define the median projection residual

$$\mathcal{M}(y; r, \ell, u) = \max(\min(y - r, y - \ell), y - u). \quad (12)$$

The equation $\mathcal{M} = 0$ returns the reference when it is feasible and the nearest bound otherwise. P-first operation is

$$\mathcal{M}(P_{\text{ac},c}; P_c^{\text{ref}}, -R_c, R_c) = 0, \quad (13a)$$

$$R_{Q,c} = \sqrt{\max(0, R_c^2 - P_{\text{ac},c}^2)}, \quad (13b)$$

$$\mathcal{M}(Q_{\text{ac},c}; Q_c^{\text{ref}}, -R_{Q,c}, R_{Q,c}) = 0, \quad (13c)$$

$$\lambda_c = 0. \quad (13d)$$

Q-first operation exchanges P and Q in (13). A finite selected generalized derivative is used when the remaining radius is zero.

Figure 2 shows the geometric distinction among the three policies for one normalized infeasible request. Magnitude limiting moves radially to the current circle. P-first retains the feasible active component before assigning the remaining reactive radius, whereas Q-first performs the dual construction. The figure is an equation-level illustration; measured production results are reported separately in Section VI.

All power-port priorities close converter energy balance through

$$P_{\text{dc},c} + P_{\text{ac},c} + P_c^{\text{loss}}(P_{\text{ac},c}, V_{\text{dc},c}) = 0, \quad (14)$$

with analytic derivatives from the selected production loss model.

E. Grid-forming Norton equation family

Let

$$\mathcal{V}_c = V_c e^{j\theta_c}, \quad E_c = E_{r,c} + jE_{i,c}, \quad Z_{v,c} = R_{v,c} + jX_{v,c}. \quad (15)$$

The converter current and terminal power are

$$I_c = \frac{E_c - \mathcal{V}_c}{Z_{v,c}}, \quad S_{\text{ac},c} = \mathcal{V}_c \overline{I_c} = P_{\text{ac},c} + jQ_{\text{ac},c}. \quad (16)$$

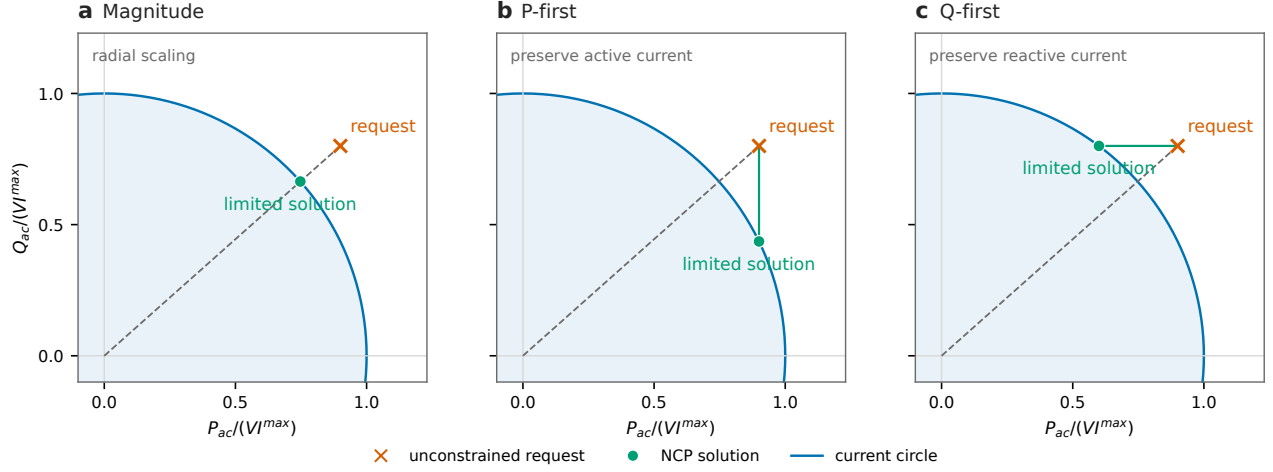


Fig. 2. Normalized current-circle geometry for the common infeasible request $(P^{\text{ref}}, Q^{\text{ref}})/(VI^{\text{max}}) = (0.9, 0.8)$. The orange cross is the unconstrained command and the green point is the NCP solution. (a) Magnitude priority scales both components radially. (b) P-first preserves active current and clips reactive current to the remaining radius. (c) Q-first preserves reactive current and clips active current.

Thus the first two local equations enforce the real and imaginary parts of (16), and (14) supplies the third.

For magnitude limiting, let E_c^0 be the authored internal-voltage reference. The remaining equations are

$$(1 + \lambda_c)(E_c - \mathcal{V}_c) - (E_c^0 - \mathcal{V}_c) = 0, \quad (17a)$$

$$\phi_{\text{FB}}(\lambda_c, (I_c^{\text{max}})^2 - |I_c|^2) = 0. \quad (17b)$$

The complex equation in (17) supplies two real rows. It can be interpreted as an implicit increase in the virtual impedance when the current circle becomes active.

For P-first and Q-first GFM limiting, I_c and the unconstrained current $I_c^0 = (E_c^0 - \mathcal{V}_c)/Z_{v,c}$ are rotated into the terminal-voltage synchronous frame. The primary current component is projected onto $[-I_c^{\text{max}}, I_c^{\text{max}}]$ using (12); the secondary component is projected onto its remaining circular radius. The unused multiplier is fixed by $\lambda_c = 0$. This construction preserves the same six rows and variables for all three priorities.

F. Model scope

The equations describe a balanced positive-sequence steady or quasi-steady operating point after limit activation. They do not represent PWM switching, the fast current-control transient, or EMT fault dynamics. A GFM terminal in a slackless island retains its terminal V/θ variables and both nodal power balances. The authored internal phasor E_c^0 breaks the global rotational null mode through (16); no artificial terminal slack is added. Multiple fixed Norton sources can coexist, with power sharing and circulating power determined by their phasors and virtual impedances.

III. Structure-Preserving Semismooth Newton Method

A. Nonsmooth residual and generalized Jacobian

Let $H : \mathbb{R}^n \rightarrow \mathbb{R}^n$ denote the complete residual in (6). Smooth ac/dc network equations and converter

loss functions are combined with the locally Lipschitz Fischer–Burmeister and median maps. The resulting H is differentiable away from switching ties and semismooth at the ties under the standard NCP construction [6], [8].

At iteration k , one element

$$J_k \in \partial_B H(w_k) \quad (18)$$

is assembled analytically. The network part follows the production convention $J_{\text{net}} = \partial(\text{calculated} - \text{specified})/\partial x$ with right-hand side equal to the network mismatch. Local converter rows store the conventional derivative of their zero residual and use right-hand side $-H_c$. This sign split is resolved during assembly into one linear system

$$J_k d_k = -H(w_k). \quad (19)$$

The exact Fischer–Burmeister map is not regularized by a hidden epsilon. At the biactive origin, the implementation selects the symmetric Clarke element

$$\left(\frac{1}{\sqrt{2}} - 1, \frac{1}{\sqrt{2}} - 1 \right) \in \partial_C \phi_{\text{FB}}(0, 0). \quad (20)$$

For the median map, the active branch determines the derivative; equal branches use an equal-weight finite selection. This policy makes the selected matrix deterministic while retaining the fixed equation layout.

B. Optional smooth continuation

Degenerate intersections may cause an abrupt change in the selected generalized Jacobian. The optional continuation replaces (11) with the Kanzow smoothing

$$\phi_\mu(a, b) = \sqrt{a^2 + b^2 + 2\mu^2} - a - b, \quad \mu > 0, \quad (21)$$

and replaces nested min/max operations in (12) with the Chen–Harker–Kanzow–Smale smooth composition [9]. The parameter is reduced from μ_0 to μ_{min} using a coarse/fine multiplicative schedule. After each reduction,

Input: Hybrid system data, initial state w_0 ,
 tolerances, μ_0
 Build the network and six-state-per-VSC sparse
 pattern once;
 Analyze the full and admissible reduced patterns
 once;
 for $k = 0, 1, \dots, k_{\max} - 1$ do
 | Evaluate exact or μ_k -smoothed $H(w_k)$;
 | if the exact target residual satisfies the
 | | stopping test then
 | | | return certified root;
 | end
 | Assemble one selected analytic generalized
 | | Jacobian J_k ;
 | Attempt the certified local-block Schur step;
 | if a Schur certificate fails then
 | | solve the complete sparse system;
 | end
 | Apply scaling and the globalized step
 | | acceptance rule;
 | if the smooth residual has decreased sufficiently
 | | and $\mu_k > \mu_{\min}$ then
 | | | reduce μ_k and reassemble at the accepted
 | | | state;
 | end
 end
 Return an explicit nonconvergence or infeasibility
 diagnosis;
 Algorithm 1: Structure-preserving semismooth Newton
 solve

the residual and Jacobian are reassembled at the unchanged state. Only matrix values change; variable order and sparse coordinates do not.

The reported operating point is always re-evaluated with $\mu = 0$. Thus a small smooth surrogate residual is not reported as an exact complementarity certificate. Smooth continuation is a globalization aid, not a change to the target physical model.

C. Globalized iteration

The production method combines the local semismooth step with residual and variable scaling, a nonmonotone line search, and fail-closed fallbacks. Let D_F and D_w denote residual and variable scaling matrices. The scaled linear system is

$$(D_F J_k D_w^{-1}) \hat{d}_k = -D_F H(w_k), \quad d_k = D_w^{-1} \hat{d}_k. \quad (22)$$

The line search accepts $w_{k+1} = w_k + \alpha_k d_k$ when the merit function $\Psi(w) = \frac{1}{2} \|D_F H(w)\|_2^2$ satisfies a non-monotone sufficient-decrease test. Trust-region, pseudo-transient, active-set-seeded, and homotopy paths remain external global safeguards; none changes the fixed local VSC state.

D. Structure preservation

Each admitted converter always contributes six rows and six columns. A limit activation, priority tie, or droop

saturation modifies residual and Jacobian values but not coordinates. Therefore the matrix memory layout, device-to-row mapping, symbolic analysis, and result index remain stable. This property removes repeated structural allocation and is also the prerequisite for the local Schur method in Section IV.

The fixed pattern does not imply that every numerical factorization can reuse the same pivots. Numeric-only KLU refactorization is attempted only after a previous factorization and is guarded by a normwise backward-error test. A failed refactorization or an inaccurate solution triggers fresh pivoting.

IV. Exact Local VSC-Block Schur Elimination

A. Block structure

Order the Newton variables so that all network variables precede the local VSC states. The selected generalized Jacobian and right-hand side are

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} d_x \\ d_z \end{bmatrix} = \begin{bmatrix} r_x \\ r_z \end{bmatrix}, \quad D = \text{diag}(D_1, \dots, D_m), \quad (23)$$

where $D_c \in \mathbb{R}^{6 \times 6}$. Converter c couples only to its terminal angle, terminal voltage magnitude, and dc voltage. Conversely, its explicit power variables enter only the terminal ac-P, ac-Q, and dc-P network rows.

When every D_c is nonsingular, block Gaussian elimination gives

$$S = A - BD^{-1}C, \quad (24a)$$

$$\hat{r}_x = r_x - BD^{-1}r_z, \quad (24b)$$

$$Sd_x = \hat{r}_x, \quad (24c)$$

$$d_z = D^{-1}(r_z - Cd_x). \quad (24d)$$

The implementation does not form D_c^{-1} . One dense full-pivoting factorization of each 6×6 block solves the four right-hand sides $[C_c, r_c]$. The resulting 3×3 update is scattered into precomputed coordinates of the reduced sparse matrix.

Proposition 1 (Algebraic equivalence). Suppose every D_c and the Schur complement S in (24) are nonsingular. Then the reduced solve and back substitution produce the unique solution of the complete system (23).

Proof. The second block row gives $d_z = D^{-1}(r_z - Cd_x)$. Substitution into the first block row yields $Sd_x = \hat{r}_x$. Nonsingularity gives a unique d_x and then a unique d_z . Direct substitution verifies both block rows [10]. $\square$

B. Fixed reduced pattern

The network submatrix A retains its ordinary sparse coordinates. Each local block has at most three coupled network rows and three coupled network columns, so $BD^{-1}C$ modifies at most nine network coordinates per converter. Those coordinates are inserted into the reduced pattern during initialization, even when their current numerical value is zero. Consequently, switching activity does not require a new symbolic analysis.

For local state size $s = 6$ and network coupling width $c \leq 3$, the dense local work is

$$O(ms^3 + ms^2(c+1)) = O(m). \quad (25)$$

The reduced dimension is n_x rather than $n_x + 6m$. In the complete fixed pattern, one converter contributes 36 local entries, 18 local-to-network entries, and three network-to-local entries. The reduced update adds at most nine coordinates, so

$$\text{nnz}(J) - \text{nnz}(S) \geq 48m. \quad (26)$$

This is an input-pattern result, not a universal sparse-LU fill theorem. Fill, flop count, cache behavior, and ordering remain network dependent [11] and are measured experimentally.

C. Numerical certificates and fallback

Every local factorization reports a reciprocal condition estimate ρ_c . A Schur attempt is rejected if a block is nonfinite or

$$\rho_c \leq \rho_{\min}. \quad (27)$$

The default is $\rho_{\min} = \sqrt{\epsilon_{\text{mach}}}$, which separates the representation scale from the requested nonlinear accuracy. This is a conservative numerical admission test, not part of the physical model.

For a matrix M , right-hand side b , and candidate solution y , the normwise backward error is evaluated as

$$\eta(M, b, y) = \frac{\|b - My\|_{\infty}}{\|M\|_{\infty}\|y\|_{\infty} + \|b\|_{\infty}}. \quad (28)$$

Both the reduced solution $\eta(S, \hat{r}_x, d_x)$ and the reconstructed complete solution $\eta(J, r, d)$ must satisfy the configured tolerance. This complete-system check is essential because a stable reduced solve alone does not certify the back substitution. Any local-regularity, reduced-solve, or full-step failure returns control to the ordinary complete sparse-LU path [12].

D. Production admission

The $O(m)$ local work and certificates have a fixed latency that can dominate a small network factorization. Therefore Schur elimination is enabled by an empirical, configurable crossover rule rather than unconditionally. The current production default admits the path only when the network-only Newton dimension is at least 1000. This number is not a mathematical constant; it is a named default derived from the fixed case300/ACTIVSg2000 protocol and can be overridden through both the C++ and production HTTP interfaces. Setting it to zero forces a research ablation.

V. Regularity and Local Convergence Conditions

A. Semismoothness assumptions

The following statements concern the exact target residual H with $\mu = 0$.

Assumption 1 (Local model regularity). The smooth ac/dc network, loss, and droop functions are continuously differentiable in a neighborhood of a solution w^* . Terminal voltage magnitudes and dc voltages remain inside the domain on which their physical equations and derivatives are finite, and each admitted GFM virtual impedance is nonzero.

Assumption 2 (BD-regular solution). The exact residual H is semismooth at w^* and every matrix $J \in \partial_B H(w^*)$ is nonsingular.

The Fischer–Burmeister map and piecewise-affine median projection are strongly semismooth. Their composition with the smooth network and converter functions preserves local semismoothness under Assumption 1. At a zero remaining-current radius, the square-root radius in the priority model requires the explicitly selected finite generalized branch used by the implementation. Such a selection makes one Newton step well defined, but it does not by itself prove Assumption 2 for all neighboring selections.

Theorem 1 (Conditional local rate). Let w^* solve $H(w) = 0$. Under Assumptions 1 and 2, the exact semismooth Newton iteration (19), started sufficiently close to w^* and eventually accepting the full step, is well defined and converges superlinearly to w^* . If H is strongly semismooth at w^* , the local rate is quadratic.

Proof. The result is the standard semismooth Newton theorem applied to the square locally Lipschitz residual H . BD-regularity gives uniformly bounded inverses for the relevant B-subdifferential elements near w^* , while semismoothness makes the linearization remainder $o(\|w - w^*\|)$. Strong semismoothness improves the remainder to $O(\|w - w^*\|^2)$ [6], [8]. $\square$

B. Block nonsingularity condition

Proposition 2 (Selected-element block certificate). Consider one selected generalized-Jacobian element J partitioned as in (23). If every D_c is nonsingular, then

$$J \text{ is nonsingular} \iff S = A - BD^{-1}C \text{ is nonsingular.} \quad (29)$$

Proof. Block Gaussian elimination factors J into nonsingular unit triangular factors and $\text{diag}(S, D)$. The determinant and null-space equivalence follow immediately [10]. $\square$

Equation (29) is the theoretical basis of the runtime Schur certificate. Local D_c condition estimates, successful sparse factorization of S , and the complete backward-error test provide numerical evidence for the current selected element. They do not enumerate $\partial_B H(w^*)$ and therefore must not be reported as a proof of BD-regularity.

C. Measured local-rate diagnostics

For accepted iterates with residual norm $r_k = \|H(w_k)\|$, the implementation reports

$$q_k^{(1)} = \frac{r_{k+1}}{r_k}, \quad q_k^{(2)} = \frac{r_{k+1}}{r_k^2}. \quad (30)$$

A decreasing $q_k^{(1)}$ is consistent with superlinear behavior, while a bounded $q_k^{(2)}$ in the asymptotic regime is consistent with quadratic behavior. These finite samples are diagnostic evidence only. Damped steps, smoothing-parameter reductions, mode-degenerate iterations, and full-LU fallbacks can occur outside the final local regime and should be excluded from any empirical rate fit.

D. Smoothing and globalization boundary

The smooth continuation in (21) improves path following but does not replace Theorem 1. The final state is accepted only after evaluation of the exact $\mu = 0$ residual. Likewise, line search and fallback scheduling improve robustness but do not establish global convergence or uniqueness. Hybrid power-flow equations may possess both high- and low-voltage mathematical roots, and a distant initial point may converge to a different basin. Infeasible operating points must be reported as nonconverged rather than converted into a plausible result.

E. Numerical-policy separation

Machine epsilon is a single read-only representation constant. Nonlinear tolerances, the local *rcond* threshold, the backward-error threshold, and the Schur crossover dimension are named algorithmic policy. They are configurable and validated separately. This prevents a hardware-scale constant from being reused as a physical feasibility tolerance and prevents empirical admission settings from becoming hidden universal assumptions.

VI. Preliminary Production Results

A. Implementation and protocol

The method is implemented in the production balanced Newton path of the HySim-XJTU-HRPES C++20 codebase. All preliminary timings use an AppleClang 21 arm64 macOS Release build with KLU. The Schur benchmark parses each case once, performs two warmups, and then executes five alternating full/Schur solves. The reported times are medians. The benchmark forces Schur on case300 for research ablation even though the production admission rule excludes it.

The current KLU adapter does not report factor nonzeros or factor work. Those fields are explicitly marked unavailable; the results below report input structural nonzeros, dimensions, iterations, and measured time without fabricating fill statistics.

B. Functional and numerical verification

The registered VSC-limit suite passes 45 cases and 540 assertions. Its scope is summarized in Table I. Analytic generalized Jacobians are checked by finite differences away from switching ties, and the biactive Fischer–Burmeister origin is tested without a hidden perturbation.

The OpenDSS evidence has two levels. At the nonbinding point, OpenDSS solves an equation-equivalent two-source Thevenin circuit independently. At the binding point, OpenDSS has no native equivalent of the proposed priority-NCP control; the internal voltage obtained by HySim is frozen and the external circuit is re-solved. The latter validates the electrical circuit and terminal KCL, but it is not an independent mode-selection oracle.

The focused transient initialization filter passes four cases and 51 assertions. It verifies stable-ID transfer of internal voltage, terminal current, and terminal power before dynamic network trimming, including a slackless GFM island. The maximum accepted seed error for each quantity is 10^{-8} p.u. A later dynamic trim may move the equilibrium when the dynamic device set does not reproduce the steady-state slack dispatch, so this result is an initialization consistency certificate rather than a transient-stability claim.

C. Schur algebraic oracle

A small sparse algebraic oracle compares the Schur step with a direct complete LU solution. The relative step error is at most 10^{-10} and the reconstructed complete-system backward error is at most 10^{-12} . A singular selected local block is rejected, and an integration test forces the production solver to fall back to complete sparse LU while retaining convergence.

For the network benchmarks, Schur and full-LU solutions satisfy

$$\|\Delta V\|_\infty, \quad \|\Delta \theta\|_\infty, \quad \|\Delta V_{dc}\|_\infty \leq 10^{-8}. \quad (31)$$

This establishes numerical equivalence of the converged roots within the declared tolerance; it does not assert uniqueness of the nonlinear root.

D. Large hybrid benchmark results

The case300 benchmark contains 300 ac buses, six dc buses, and six explicit VSC blocks. ACTIVSg2000 contains 2000 ac buses, eight dc buses, and eight VSC blocks. Each case has four simultaneous binding power-port limits. These are real hybrid networks rather than pure-ac cases with timing labels added after the fact.

Figure 3 presents the measured structure, timing, and certificate outcomes before the exact values are tabulated. It makes the production crossover visible: the same reduction is detrimental on case300 but beneficial on ACTIVSg2000. It also shows that the large-case speedup is obtained while rejecting numerically inadmissible Schur steps rather than silently accepting every reduced solve.

Figure 3(a)–(b) confirms the structural predictions: six states are removed per VSC, and the observed structural

TABLE I
Current verification coverage and acceptance boundaries

Category	Covered cases	Acceptance evidence
Local equations	Magnitude, P-first, and Q-first power ports; all three GFM priorities; Vdc-droop saturation; zero-radius and biactive intersections	Local/oracle and exact complementarity residuals at most 10^{-9} ; biactive residual at most 2×10^{-15}
Analytic derivatives	Local power-port and GFM terminal/internal-voltage derivatives; smooth FB and smoothed median derivatives	Centered finite-difference discrepancy at most 5×10^{-7}
Network integration	Two simultaneous power-port limits, two simultaneous GFM limits, weak-grid roots, and deliberately infeasible weak-grid points	One fixed Jacobian-pattern build; infeasible points are not reported as converged
Cross-process consistency	Three-step quasi-steady replay, balanced OPF dispatch replay, stable-ID JSON round trip, GFM-island extraction, and transient initialization	OPF-to-PF exact certificate at most 10^{-9} ; PF-to-transient Norton seed errors at most 10^{-8}
External circuit validation	Nonbinding GFM Thevenin point and binding frozen-internal-voltage circuit in OpenDSS	Nonbinding $ \Delta V \leq 10^{-4}$ p.u.; binding native terminal KCL residual at most 10^{-6} p.u.

reductions exceed the 48m lower bound. It also demonstrates why structure preservation must be measured rather than assumed. On case300, local factorization and certificate overhead dominate the small network factorization, so forced Schur is slower. On ACTIVSg2000, the reduced factorization is large enough to offset the local work, lowering both linear and end-to-end median time.

The full/Schur median linear and wall times are respectively 0.491/0.692 ms and 2.675/3.094 ms for case300, and 31.561/27.135 ms and 47.391/41.486 ms for ACTIVSg2000. Thus the visual percentage changes in Fig. 3(c) retain their underlying absolute measurements without requiring a duplicate wide table.

For case300, all ten Schur attempts are accepted. The minimum accepted local $rcond$ is 5.48×10^{-2} and the maximum accepted full backward error is 3.63×10^{-17} . For ACTIVSg2000, 19 of 33 Schur attempts are accepted; ten are rejected by the local-regularity guard and four by the reconstructed full backward-error guard. The minimum accepted local $rcond$ is 2.19×10^{-8} , and the maximum accepted full backward error is 5.96×10^{-17} . Rejected attempts use the complete sparse-LU path rather than terminating the nonlinear solve.

E. Interpretation and present limitations

The preliminary evidence supports three current claims: the explicit NCP model remains structurally fixed across activation; the local Schur solve is algebraically equivalent when its certificate is accepted; and a measurable CPU benefit appears on the tested 2000-bus hybrid case. The evidence does not yet support a universal crossover dimension, GPU acceleration, sparse-factor fill reduction, or large-scale simultaneous binding GFM behavior. The default dimension of 1000 is therefore exposed as configurable policy and is not treated as a theorem.

The large GFM datasets currently add one nonbinding Norton GFM block to a certified MTDC operating point through continuation. They verify data/model integration and fixed structure, but not flat-start robustness or simultaneous large-scale GFM limiting. Likewise, weak-grid cases exercise difficult local roots but do not replace

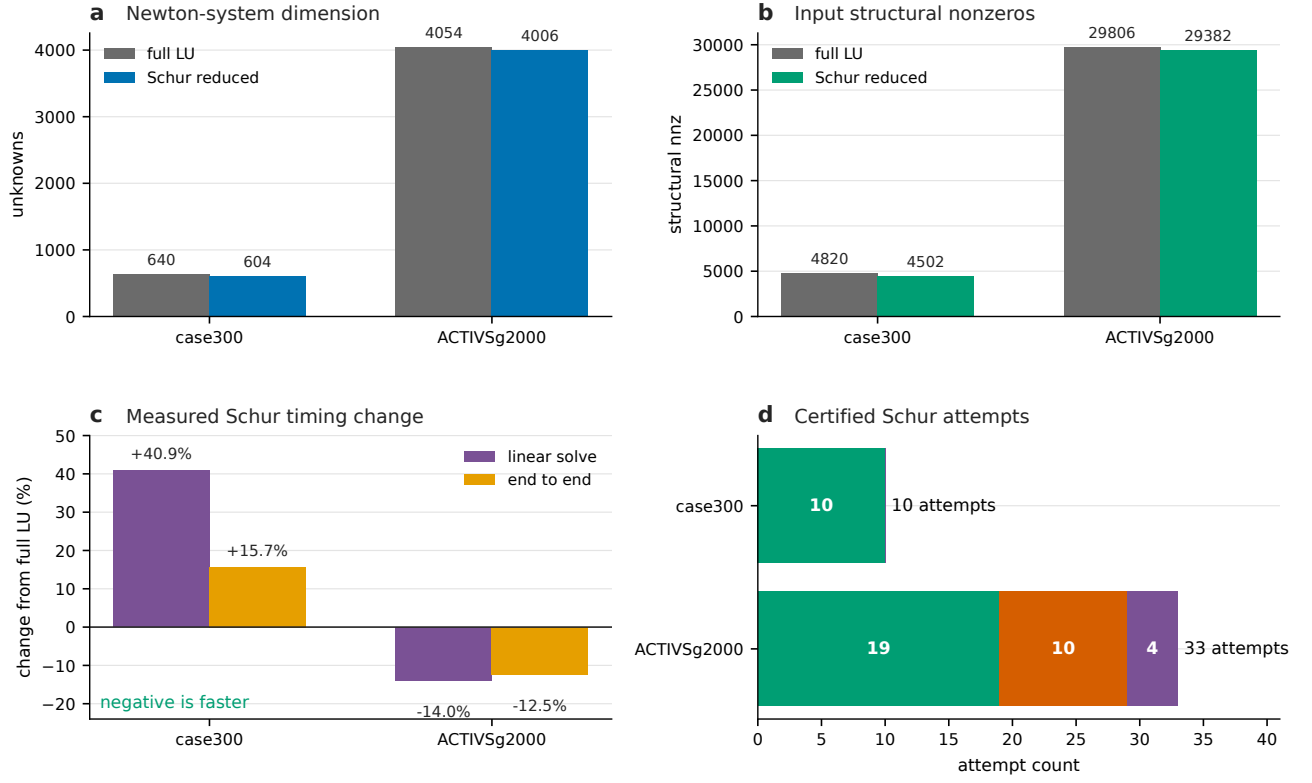
a systematic CPF continuation toward a voltage-stability nose. These experiments define the remaining work needed before the results section is submission complete.

VII. Current Scope and Next Experimental Closures

The present manuscript establishes the balanced positive-sequence model and the CPU sparse-solver contribution. It does not claim GPU assembly or GPU factorization. The large hybrid benchmarks contain simultaneous binding power-port converters, whereas their added large-scale GFM blocks are currently nonbinding continuation cases. A complete active-mode enumeration oracle, systematic continuation to the voltage-stability nose, and independent external selection of the binding GFM priority mode remain required before the experimental section is considered final. Balanced OPF currently preserves the authored GFM mode and certifies its dispatch by post-OPF power-flow replay; the priority NCP is not yet embedded in the OPF KKT system.

References

- [1] Z. Javid, T. Xue, U. Karaagac, and I. Kocar, "Unified power flow solver for hybrid AC/DC distribution networks," *IEEE Transactions on Power Delivery*, vol. 38, no. 5, pp. 3322–3332, 2023.
- [2] Q. Sun, Y. Zhang, Y. Zhou, J. Wang, and J. Wang, "Multi-dimensional holomorphic embedding three-phase power flow for AC-DC hybrid distribution systems," *IEEE Transactions on Power Systems*, vol. 39, no. 3, pp. 5310–5323, 2024.
- [3] E. Pompodakis, G. C. Karyonidis, C. S. Demoulias, and M. C. Alexiadis, "A generic power flow algorithm for unbalanced islanded hybrid AC/DC microgrids," *IEEE Transactions on Power Systems*, vol. 36, no. 2, pp. 1107–1120, 2021.
- [4] J. Rocabert, A. Luna, F. Blaabjerg, and P. Rodriguez, "Control of power converters in AC microgrids," *IEEE Transactions on Power Electronics*, vol. 27, no. 11, pp. 4734–4749, 2012.
- [5] N. Baeckeland, D. Venkatramanan, M. Kleemann, and S. Dhople, "Stationary-frame grid-forming inverter control architectures for unbalanced fault-current limiting," *IEEE Transactions on Energy Conversion*, vol. 37, no. 4, pp. 2813–2825, 2022.
- [6] F. Facchinei and J.-S. Pang, *Finite-Dimensional Variational Inequalities and Complementarity Problems*. Springer, 2003, vol. 1.



Release/KLU; 2 warmups and 5 alternating repetitions; medians reported.

Fig. 3. Production CPU/KLU Schur benchmark using the fixed two-warmup, five-repeat alternating protocol. (a) Full and reduced Newton dimensions. (b) Input structural nonzeros; factor-fill statistics are unavailable from the current adapter and are not inferred. (c) Median timing change relative to complete sparse LU, where negative values indicate acceleration. (d) Accepted and rejected Schur attempts: green denotes accepted steps, orange denotes local-regularity rejection, and purple denotes reconstructed full-step backward-error rejection. case300 is a forced research ablation; the production admission rule excludes it.

- [7] A. Fischer, “A special newton-type optimization method,” *Optimization*, vol. 24, pp. 269–284, 1992.
- [8] L. Qi and J. Sun, “A nonsmooth version of newton’s method,” *Mathematical Programming*, vol. 58, pp. 353–367, 1993.
- [9] C. Kanzow, “Some noninterior continuation methods for linear complementarity problems,” *SIAM Journal on Matrix Analysis and Applications*, vol. 17, no. 4, pp. 851–868, 1996.
- [10] G. H. Golub and C. F. V. Loan, *Matrix Computations*, 4th ed. Johns Hopkins University Press, 2013.
- [11] T. A. Davis, *Direct Methods for Sparse Linear Systems*. SIAM, 2006.
- [12] N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd ed. SIAM, 2002.